

Recorded Mode

Country: Norway

Company: Equinor

Well: 31/5-7

Field: Eos

Rig Name: West Hercules

Country: Norway

Latitude: 60° 34' 35.14" N

UWID:

Longitude: 3° 26' 36.13" E

Rig Name:

West Hercules

Block: 31/5

Rig Type:

Semi-Submersible

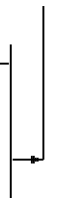
FL: Norwegian North Sea

FL1: Section: 8 1/2 in. pilot

FL2: Job no.: 19SCA0085

Rig Name: West Hercules

Log Measured From: - Drill Floor: 31.00 m
Permanent Datum: - Mean Sea Level



Ground Level: 307.00 m

Acquisition Dates: 04-Dec-2019

Other Services:

Log Interval: 389.00(m) -- 946.00(m)

Direction & Inclination (Surveys)

Index Types: Measured Depth

Directional Drilling

Index Scales: 1:500

Annular Pressure While Drilling

Depth Source: Driller's Depth

SonicScope

Depth Sensor: 3rd Party Depth

Print Type: Final

Spud Date: 02-Dec-2019



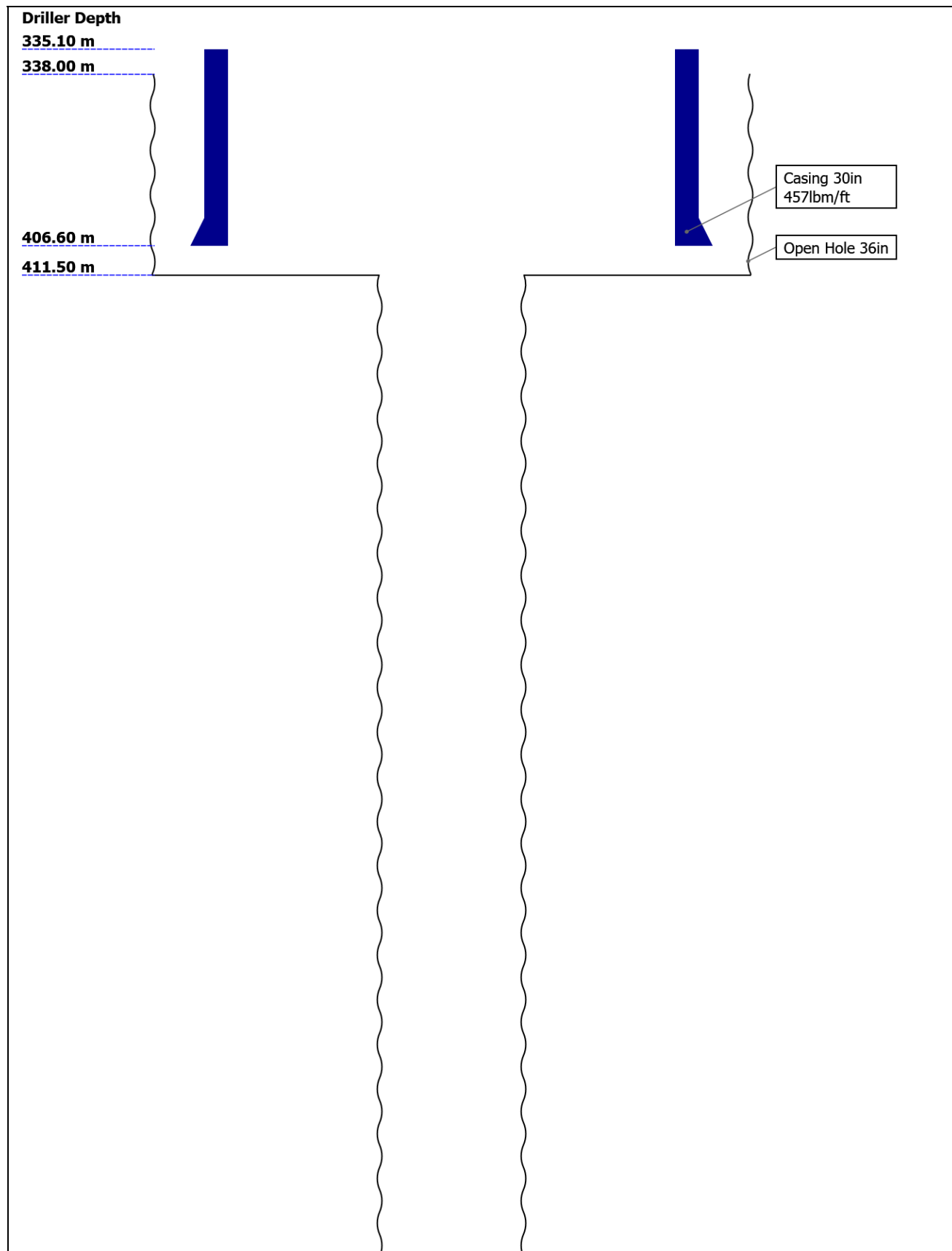
Disclaimer

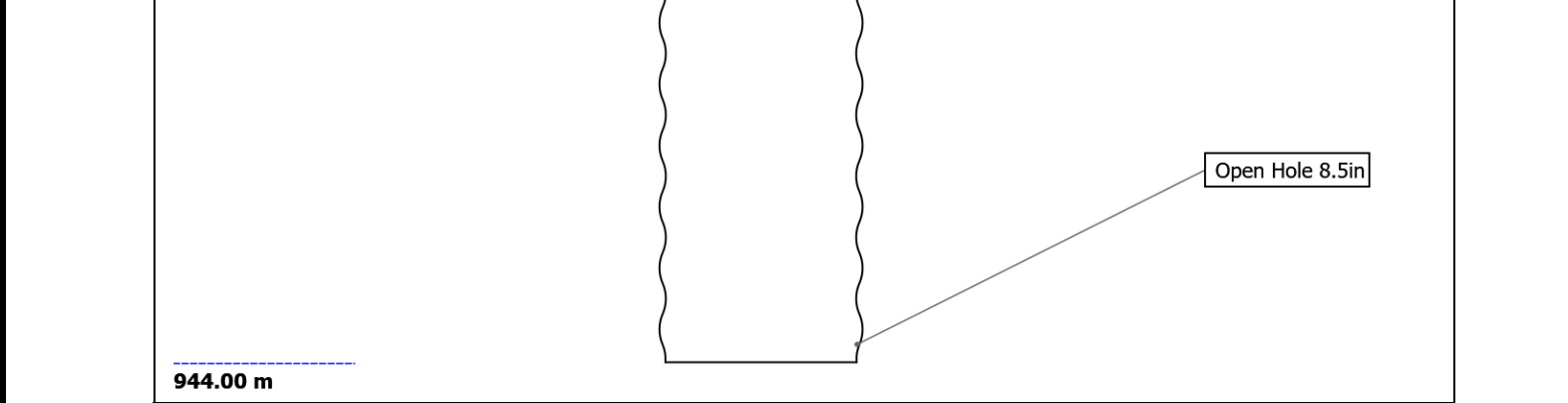
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Well Sketch





Borehole Size/Casing Record

Bit						
Bit Size (in)	36	8.5				
Top Driller (m)	338	411.5				
Bottom Driller (m)	411.5	944				
Casing						
Size (in)	30					
Weight (lbm/ft)	457					
Inner Diameter (in)	27.067					
Grade	X56					
Top Driller (m)	335.1					
Bottom Driller (m)	406.6					

Operational Run Summary

Parameter (unit)	Run 2					
Date Log Started	04-Dec-2019					
Time Log Started	00:53:56					
Date Log Finished	04-Dec-2019					
Time Log Finished	20:23:38					
Bit Size (in)	8.500					
Bit Start Depth (m)	411.50					
Bit Stop Depth (m)	944.00					
Top Log Interval (m)	406.80					
Bottom Log Interval (m)	936.20					
Max Hole Deviation (deg)	0.18					
Azimuth of Max Deviation (deg)	108.58					
Logging Unit Number	A615					
Logging Unit Location	Service Office					
Recorded By	J.Johansen					
Witnessed By	R.Nordstokke					
Service Order Number	19SCA0085					

Borehole Fluids

Parameter(unit)	Run 2					
Fluid Type	Water					
Fluid Name	Seawater					
Max Recorded Temperatures (degC)	16					
Source of Sample	Active Tank					
Salinity (ppm)	33579.09					
Density (g/cm3)	1.03					
Funnel Viscosity (s)						
Fluid Loss (cm3)						
PH						
Source RMF						
RMC	Pressed					
RM @ Meas Temp (ohm.m@degC)	0.19 @ 23.9					
RMF @ Meas Temp (ohm.m@degC)	0.19 @ 23.9					
RMC @ Meas Temp (ohm.m@degC)						
RM @ BHT (ohm.m@degC)	0.23 @ 16					
RMF @ BHT (ohm.m@degC)	0.23 @ 16					
RMC @ BHT (ohm.m@degC)	NaN @ 16					
Total Solid (%)						
High Gravity Solids (%)						

Remarks and Equipment Summary

Run 2: Toolstring				Run 2: Remarks	
Equip name SONICSCOPE6:H 0121 MP6DC:H0121 Upper Stabilizer: 00 164-01 Lower Stabilizer: 00 370-2-2 Upper Extender MP6UC:8019 MP6LC:8019 Lower Extender: RF E-11 XOS6:78001	Length 32.32		MP name SonicScope Offset	All depths are referenced to driller's depth. See EOWR for depth tracking details.	
				All data from tool memory. All data acquired while drilling.	
				Gamma Ray measurement is environmentally corrected for mud weight, bit size, collar thickness and neutron activation.	
				Resistivity measurements are borehole compensated and environmentally corrected for bit size, mud resistivity and temperature.	
				Bulk Density is compensated for tool standoff/mud cake.	
				Neutron Porosity is calculated with limestone matrix and is environmentally corrected for bit size, mud weight, temperature, pressure and mud salinity.	
				Small intervals of missing Neutron Porosity data due to the practice of quickly resuming drilling after connections, before the tool has sufficient time to start PNG firing.	
				Interval of missing data at 532.6 m bit depth due to low flow when resuming drilling after connection.	
				Run 2 and 8 1/2 in. pilot section TD at 944 m.	

TeleScope

D&I 18.11

Vibration **17.11**

ROP **15.76**

EcoScope

PNG Monitor **11.56**

Sigma **11.36**

Spectroscopy 11.36

Neutron	11.06
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Reactivity 10.91

Resistivity 10.81

UltraSonic 9.39

Density 8.97

Inclinometer 8.29

ROP 8 06

ROP	8.00
ROP	5.00

Pressure 7.93

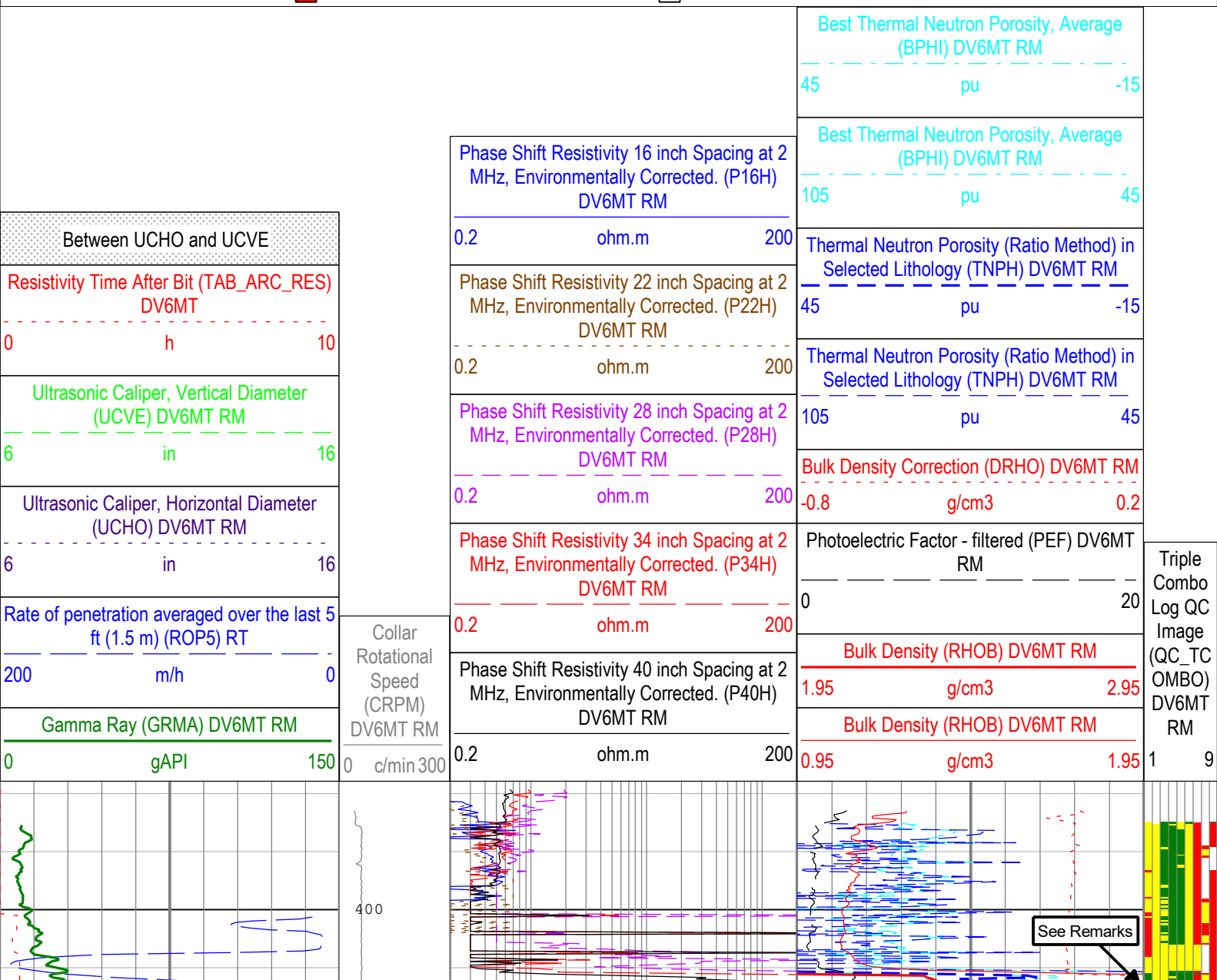
GR **7.77**

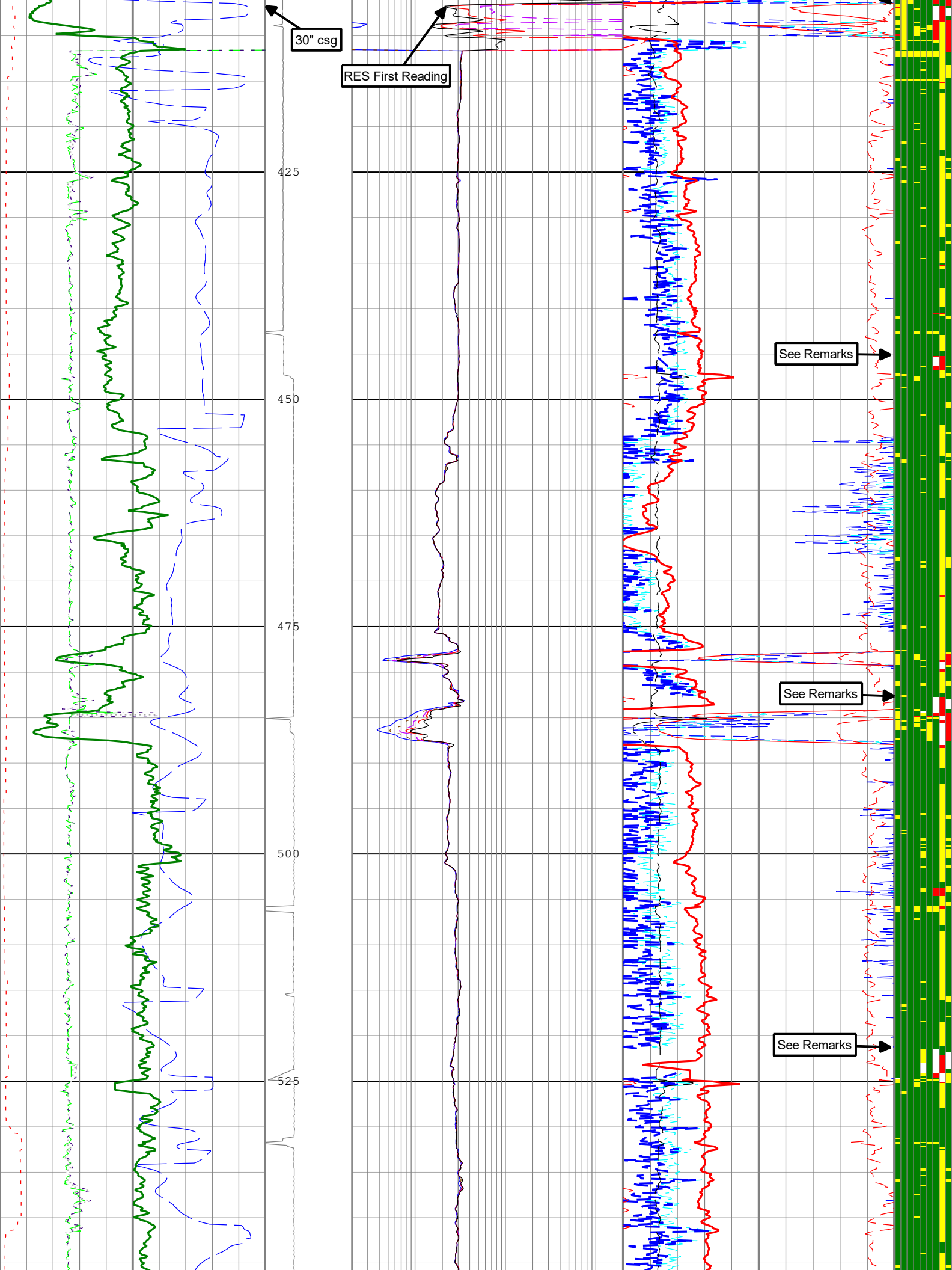
Lengths are in m
Maximum Outer Diameter = 8.500 in
Line: Sensor Location, Value: Gating Offset
All measurements are relative to TOOL ZERO

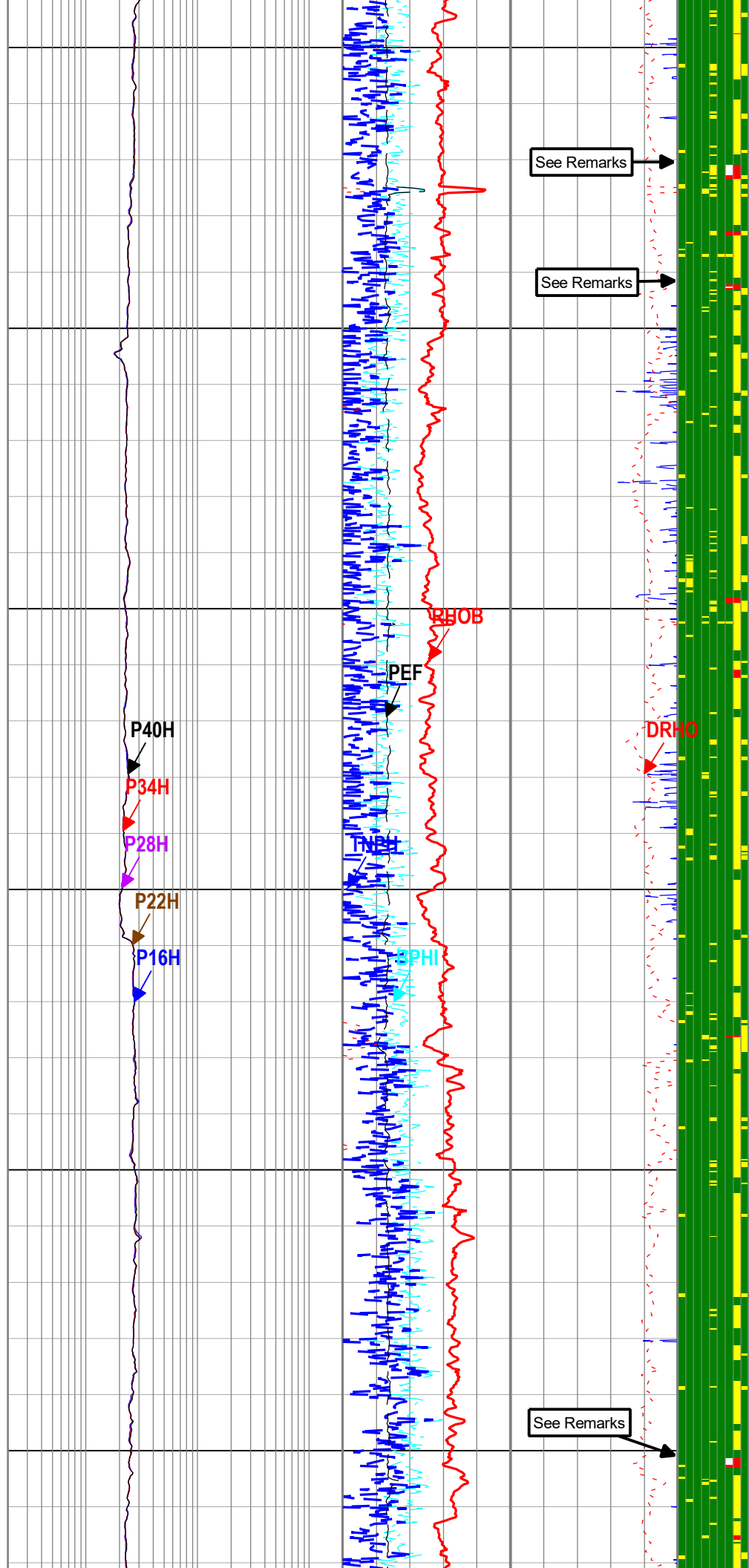
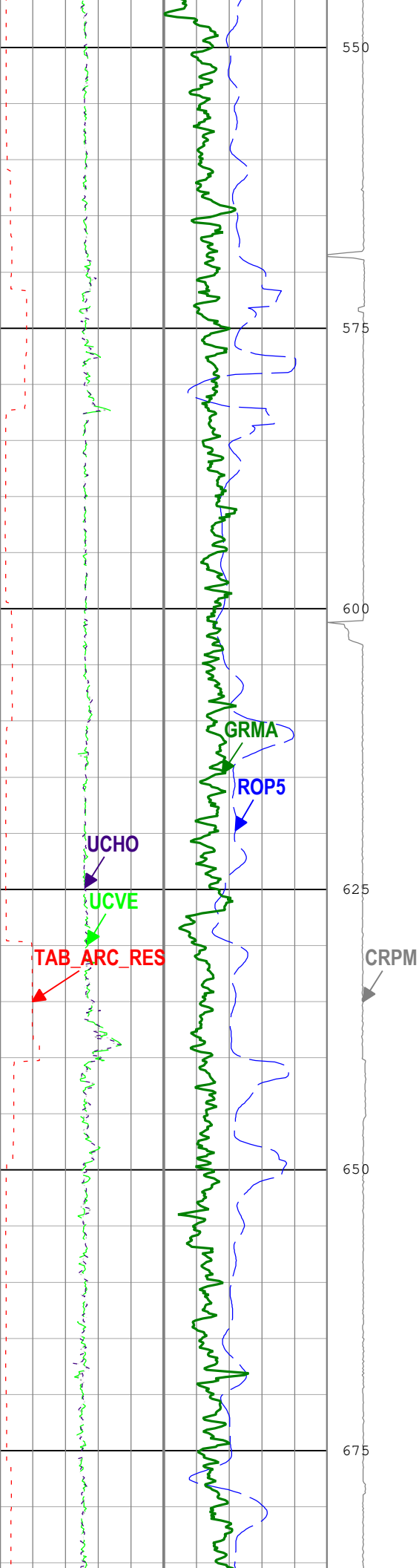
Run 2: Drilling:S019

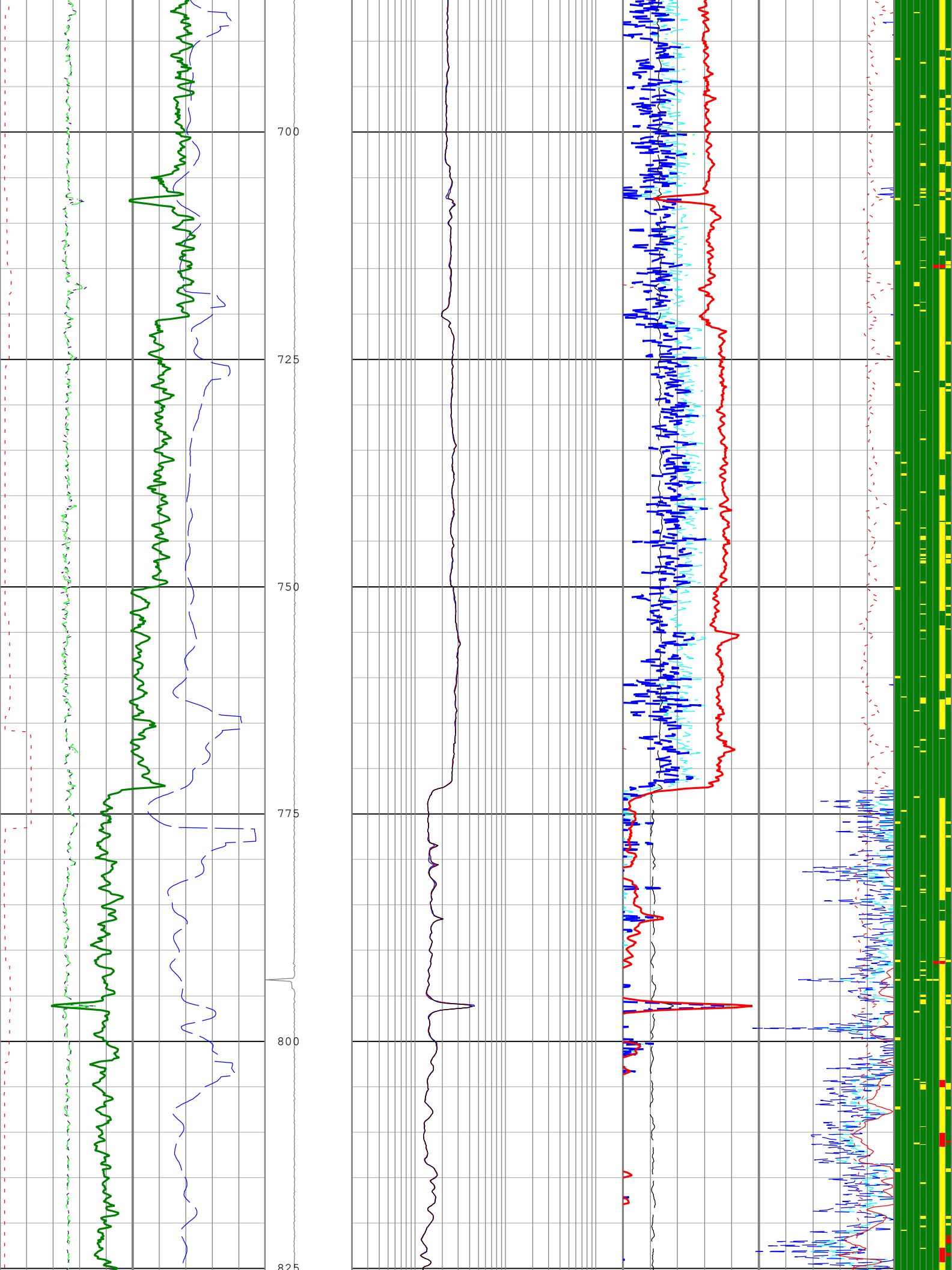
Triple Combo Log QC Image (QC_TCOMBO) DV6MT RM

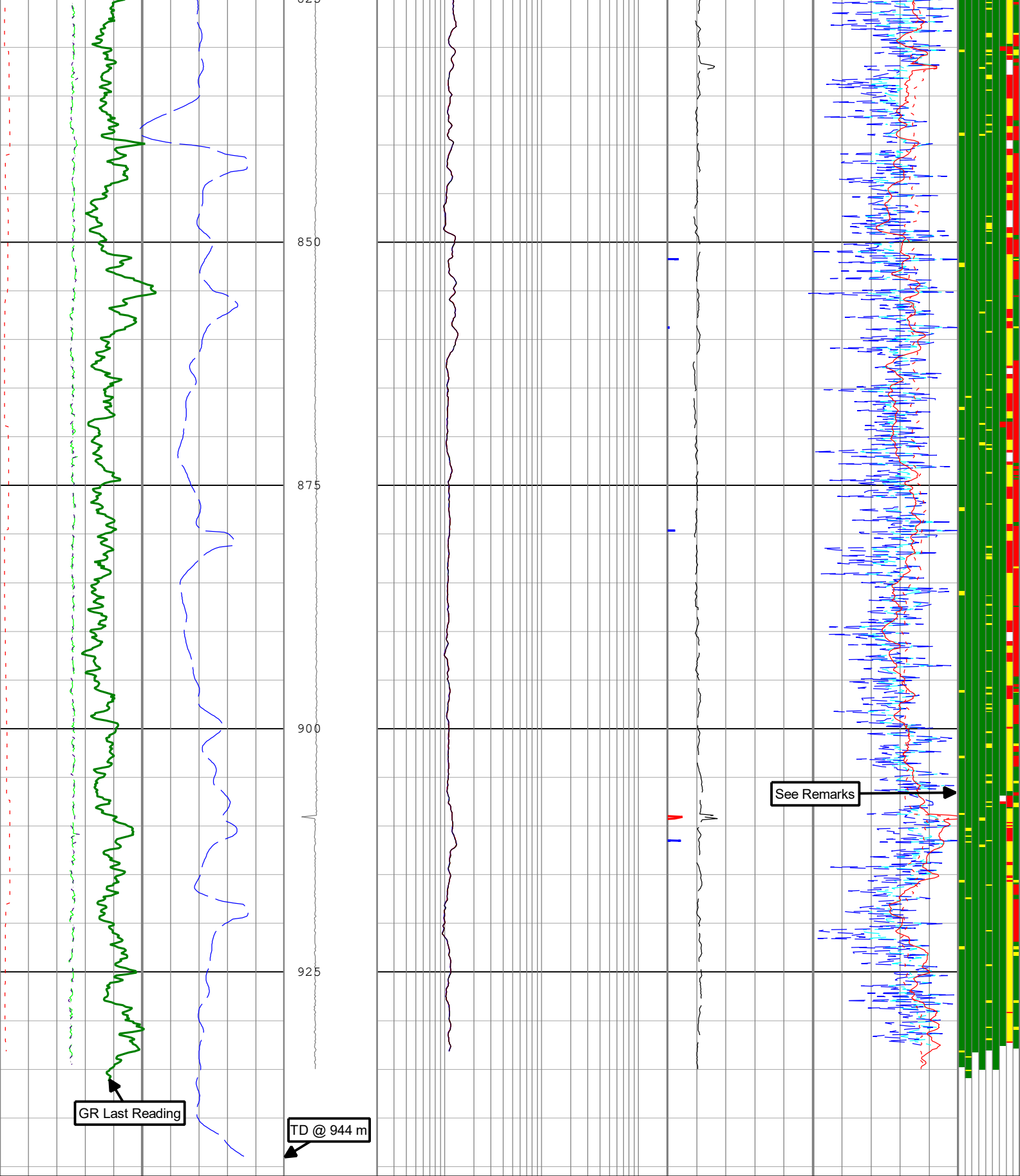
1 - Ultrasonic Caliper Quality - :	 Caliper Reliable	 Caliper Questionable
	 Caliper properties out of tolerance	 ABSENT
2 - GR Quality - :	 Gamma Ray Reliable	 Gamma Ray Questionable
	 Gamma Ray properties out of tolerance	 ABSENT
3 - Resistivity Quality - :	 Resistivity Reliable	 Resistivity Questionable
	 Resistivity properties out of tolerance	 ABSENT
4 - Density Quality - :	 Density Reliable	 Density Questionable
	 Density properties out of tolerance	 ABSENT
5 - Porosity Quality - :	 Porosity Reliable	 Porosity Questionable
	 Porosity properties out of tolerance	 ABSENT
6 - Photoelectric Factor Quality - :	 Photoelectric Factor Reliable	 Photoelectric Factor Questionable
	 Photoelectric Factor out of tolerance	 Photoelectric Factor ABSENT
7 - Sigma Quality - :	 Sigma Reliable	 Sigma Questionable
	 Sigma out of tolerance	 Sigma ABSENT
8 - Spectroscopy Quality - :	 Spectroscopy Reliable	 Spectroscopy Questionable
	 Spectroscopy out of tolerance	 Spectroscopy ABSENT
9 - RHON Quality - :	 RHON Reliable	 RHON Questionable
	 RHON out of tolerance	 RHON ABSENT











See Remarks

GR Last Reading

TD @ 944 m

Between UCHO and UCVE		Collar Rotational Speed (CRPM) DV6MT RM	Phase Shift Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected. (P16H) DV6MT RM		Best Thermal Neutron Porosity, Average (BPHI) DV6MT RM		Triple Combo Log QC Image (QC_TC OMBO) DV6MT RM
Resistivity Time After Bit (TAB_ARC_RES) DV6MT			0.2 ohm.m 200		45 pu -15		
0 h 10		0 c/min 300	Phase Shift Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected. (P22H) DV6MT RM		Best Thermal Neutron Porosity, Average (BPHI) DV6MT RM		1
Ultrasonic Caliper, Vertical Diameter (UCVE) DV6MT RM					105 pu 45		

6	in	16	0.2	ohm.m	200	Thermal Neutron Porosity (Ratio Method) in Selected Lithology (TNPH) DV6MT RM
Ultrasonic Caliper, Horizontal Diameter (UCHO) DV6MT RM			Phase Shift Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected. (P28H) DV6MT RM			45 pu -15
6	in	16	0.2	ohm.m	200	Thermal Neutron Porosity (Ratio Method) in Selected Lithology (TNPH) DV6MT RM
Rate of penetration averaged over the last 5 ft (1.5 m) (ROP5) RT			Phase Shift Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected. (P34H) DV6MT RM			105 pu 45
200	m/h	0	0.2	ohm.m	200	Bulk Density Correction (DRHO) DV6MT RM
Gamma Ray (GRMA) DV6MT RM			Phase Shift Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (P40H) DV6MT RM			-0.8 g/cm3 0.2
0	gAPI	150	0.2	ohm.m	200	Photoelectric Factor - filtered (PEF) DV6MT RM
						0 20
						Bulk Density (RHOB) DV6MT RM
						1.95 g/cm3 2.95
						Bulk Density (RHOB) DV6MT RM
						0.95 g/cm3 1.95

Triple Combo Log QC Image (QC_TCOMBO) DV6MT RM		
1 - Ultrasonic Caliper Quality - :	Caliper Reliable Caliper properties out of tolerance	Caliper Questionable ABSENT
2 - GR Quality - :	Gamma Ray Reliable Gamma Ray properties out of tolerance	Gamma Ray Questionable ABSENT
3 - Resistivity Quality - :	Resistivity Reliable Resistivity properties out of tolerance	Resistivity Questionable ABSENT
4 - Density Quality - :	Density Reliable Density properties out of tolerance	Density Questionable ABSENT
5 - Porosity Quality - :	Porosity Reliable Porosity properties out of tolerance	Porosity Questionable ABSENT
6 - Photoelectric Factor Quality - :	Photoelectric Factor Reliable Photoelectric Factor out of tolerance	Photoelectric Factor Questionable Photoelectric Factor ABSENT
7 - Sigma Quality - :	Sigma Reliable Sigma out of tolerance	Sigma Questionable Sigma ABSENT
8 - Spectroscopy Quality - :	Spectroscopy Reliable Spectroscopy out of tolerance	Spectroscopy Questionable Spectroscopy ABSENT
9 - RHON Quality - :	RHON Reliable RHON out of tolerance	RHON Questionable RHON ABSENT

Description: ECO Density Neutron Resistivity
Format: Log (ECO Density Neutron Resistivity_2)
Index Scale: 1:500
Index Unit: m
Index Type: Measured
Depth
Creation Date: 10-Dec-2019 14:35:10

Channel Processing Parameters				
Run 2: Parameters				
Parameter	Description	Tool	Value	Unit
ABNT	Abnormal Transmitter Indicator	DV6MT	NO_TX_FAILED	
ACT_STAB_SIZE	Actual Stabilizer Size	DV6MT	8.25	in
BH_COMPUTE	Borehole Effect Computation Option	DV6MT	No	
BHK	Drilling Fluid Potassium Concentration	Borehole	0	%
BHT	Bottom Hole Temperature	Borehole	16	degC
BS	Bit Size	DNMSESSION	Depth Zoned	in
BSAL	Borehole Salinity	Borehole	33579.09	ppm
CALI_SEL_GR	Hole-Size Correction Source for Gamma-Ray Processing	DV6MT	GCSE	
CALI_SEL_NEU	Hole-Size Correction Source for Neutron Processing	DV6MT	GCSE	

CAL_SEL_NGD	Hole-Size Correction Source for Neutron Gamma Density Processing	DV6MT	Ultrasonic	
CAL_SEL_SIG	Hole-Size Correction Source for Sigma Processing	DV6MT	GCSE	
CHI_RM	Caliper High Limit from BS (RM)	DV6MT	10	in
CLO_RM	Caliper Low Limit from BS (RM)	DV6MT	-5	in
DEPTH_SEL	Depth Selection Parameter	DNMSESSION	Driller's Depth	
DFD	Drilling Fluid Density	Borehole	1.03	g/cm3
DFT_CATEGORY	Drilling Fluid Type	Borehole	Water	
DTMD	Borehole Fluid Slowness	Borehole	248	us/ft
DTMUD_DH	Delta-T for Mud Downhole	DV6MT	190	us/ft
ERRCT	Percentage Error Cutoff	DV6MT	4.5	%
FSAL	Formation Salinity	Borehole	6126.75	ppm
GCSE_RM	Generalized Caliper Selection for DnM recorded mode	Borehole	BS	
GR_O2COR_OPT	Enable Gamma Ray Oxygen Activation Correction	DV6MT	Yes	
GRSE_RM	Generalized Mud Resistivity Selection for Recorded Mode	Borehole	REMS(RM)	
GRSH	Gamma Ray Shale	DV6MT	1000	gAPI
GTSE_RM	Generalized Temperature Selection for Recorded Mode	Borehole	DHAT(RM)	
HIGH_BLEND	High Resistivity Threshold for Blending	DV6MT	2	ohm.m
INVAS_COMPUTE	Invasion Computation Option	DV6MT	No	
LOW_BLEND	Low Resistivity Threshold for Blending	DV6MT	1	ohm.m
MATR	Rock Matrix for Neutron Porosity Corrections	Borehole	LIMESTONE	
MST	Mud Sample Temperature	Borehole	23.9	degC
MSWS	ARCWizard Model Switch Window Size	DV6MT	152.4	cm
MULTIEFFECT_COMPUTE	Multi-effect Computation Option	DV6MT	No	
NEU_DCOR_OPT	Density Correction Source for Neutron Processing	DV6MT	Average	
NEU_FTUBE_OPT	Far Thermal Tube Selection	DV6MT	Both	
NOM_STAB_SIZE	Nominal Stabilizer Size	DV6MT	8.25	in
OACF	O2 Activation Correction Factor (RM)	DV6MT	8	
NEU_PRESCOR_OPT	Pressure Correction Source for Neutron Processing	DV6MT	Annular	
PRTD	ARCWizard Preferred Resistivty Log for Rt Display while Multi-Effects	DV6MT	P34B	
RMS	Resistivity of Mud Sample	Borehole	0.19	ohm.m
RUN_TYPE	Run Duration switch	DV6MT	NORMAL	
SIG_PCOR_OPT	Porosity Correction Source for Sigma Processing	DV6MT	Best	
SPEC_SIG_COR	Apply Environment Correction (Sigma)	DV6MT	Yes	
SPL_CLAY_MODEL	SpectroLith Clay Model	DV6MT	Sub-Arkose	
SPL_COAL_OPT	SpectroLith Coal Processing Option	DV6MT	None	
SPL_FILT_LEN	Filter Length for Relative Yields	DV6MT	5	
SPL_SULFUR_MIN	SpectroLith Sulfur Mineral Option	DV6MT	Pyrite	
STOH	Top of Hole Sector	DV6MT	SECTOR_0	
T1WM	ARCWizard Weight Multiplier to Measurements for Transmitter 1	DV6MT	1	
T2WM	ARCWizard Weight Multiplier to Measurements for Transmitter 2	DV6MT	1	
T3WM	ARCWizard Weight Multiplier to Measurements for Transmitter 3	DV6MT	1	
T4WM	ARCWizard Weight Multiplier to Measurements for Transmitter 4	DV6MT	1	
T5WM	ARCWizard Weight Multiplier to Measurements for Transmitter 5	DV6MT	1	
ATMP_ARC	ARC Temperature Selection	DV6MT	Annular	
NEU_TEMPCOR_OPT	Temperature Correction Source for Neutron Processing	DV6MT	GTSE	
UNIFORM_COMPUTE	Uniform Rock Computation Option	DV6MT	No	

Depth Zone Parameters

Parameter	Value	Start (m)	Stop (m)
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BS	36	389	411.5
BS	8.5	411.5	944
All depth are actual.			

Tool Control Parameters

Run 2: Parameters

Parameter	Description	Tool	Value	Unit
OFFBTM_TH	Threshold for deciding whether the bit is off bottom	DNMSESSION	0	m

Calibration Report

DV6MT (EcoScope Integrated Logging-While-Drilling Tool) Calibration - Run 2

Primary Equipment :			
Electronics Chassis	DVME	1801	
Auxiliary Equipment :			
Drill Collar	DVMC	2855	
Gamma Gamma Logging Source	GGLS	A2609	
8 1/4 inch Stabilizer	DVMS	3811	

Resistivity Calibration -

Master (Time Frame File):		08:10:30 12-Nov-2019					
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Attenuation T1 at 2 MHz	dB	Master	9.0	7.0	8.4	11.0	
Attenuation T2 at 2 MHz	dB	Master	6.0	4.0	6.0	8.0	
Attenuation T3 at 2 MHz	dB	Master	5.5	3.5	5.0	7.5	
Attenuation T4 at 2 MHz	dB	Master	4.5	2.5	4.4	6.5	
Attenuation T5 at 2 MHz	dB	Master	4.0	2.0	3.6	6.0	
Phase Shift T1 at 2 MHz	deg	Master	0.1000	-4.0000	0.0545	4.0000	
Phase Shift T2 at 2 MHz	deg	Master	0.1000	-4.0000	-0.1075	4.0000	
Phase Shift T3 at 2 MHz	deg	Master	0.1000	-4.0000	0.0315	4.0000	
Phase Shift T4 at 2 MHz	deg	Master	0.1000	-4.0000	-0.1082	4.0000	
Phase Shift T5 at 2 MHz	deg	Master	0.1000	-4.0000	0.0531	4.0000	
Attenuation T1 at 400 KHz	dB	Master	9.0	7.0	8.4	11.0	
Attenuation T2 at 400 KHz	dB	Master	6.0	4.0	6.0	8.0	
Attenuation T3 at 400 KHz	dB	Master	5.5	3.5	5.0	7.5	
Attenuation T4 at 400 KHz	dB	Master	4.5	2.5	4.4	6.5	
Attenuation T5 at 400 KHz	dB	Master	4.0	2.0	3.6	6.0	
Phase Shift T1 at 400 KHz	deg	Master	0.1000	-4.0000	-0.6019	4.0000	
Phase Shift T2 at 400 KHz	deg	Master	0.1000	-4.0000	0.5826	4.0000	
Phase Shift T3 at 400 KHz	deg	Master	0.1000	-4.0000	-0.5729	4.0000	
Phase Shift T4 at 400 KHz	deg	Master	0.1000	-4.0000	0.5946	4.0000	
Phase Shift T5 at 400 KHz	deg	Master	0.1000	-4.0000	-0.5526	4.0000	

Gamma-Gamma-Density Calibration -

Master (Time Frame File):		18:46:21 12-Nov-2019					
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
LS window 3 - Background	1/s	Master	70.0000	50.0000	51.7165	90.0000	
LS window 3 - Al	1/s	Master	575.0000	350.0000	468.3680	800.0000	
LS window 3 - Mg	1/s	Master	3350.0000	2200.0000	2775.9300	4500.0000	
Long spacing water density	g/cm3	Master	1.044	1.028	1.050	1.061	
SS window 1 - Background	1/s	Master	75.0000	50.0000	58.0659	100.0000	
SS window 1 - Al	1/s	Master	3550.0000	2300.0000	2889.7600	4800.0000	
SS window 1 - Mg	1/s	Master	6830.0000	4560.0000	5822.0200	9100.0000	
SS window 3 - Background	1/s	Master	370.000	270.000	279.492	470.000	
SS window 3 - Al	1/s	Master	11550.000	7600.000	8992.490	15500.000	
SS window 3 - Mg	1/s	Master	16700.000	11100.000	13157.900	22300.000	
Short spacing water density	g/cm3	Master	1.080	1.045	1.097	1.115	

Neutron Porosity Calibration -

Master (Time Frame File):		16:23:58 12-Nov-2019					
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Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit		
Porosity Calibration Pass=1/Fail=0		Master	1.0	0.5	1.0	1.5		
Far 1 Gain		Master	1.000	0.700	1.047	1.300		
Far 1 Offset	1/s	Master	0.001	-3.000	0.353	3.000		
Far 2 Gain		Master	1.000	0.700	0.992	1.300		
Far 2 Offset	1/s	Master	0.001	-3.000	0.422	3.000		
Thermal Near Gain		Master	1.001	0.700	1.065	1.300		
Thermal Near Offset	1/s	Master	0.001	-500.000	-142.817	500.000		
Spectroscopy Calibration -								
Master (Time Frame File): 16:23:58 12-Nov-2019								
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit		
FECOR Gain		Master	55.0	29.0	55.0	81.0		
FECOR Offset		Master	0	-7.0	0.0	7.0		

Company:

Equinor

Well:

31/5-7

Field:

Eos

Rig Name:

West Hercules

Country:

Norway



EcoScope Service

Schlumberger

Recorded Mode